Shaun Richter

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Education

University of Iowa January 2025 - December 2025
M.S. in Electrical and Computer Engineering *GPA: 4.00*

University of Iowa August 2021 - December 2024
Honors B.S.E. in Computer Science and Engineering, Highest Distinction *GPA 4.06*

- **Coursework**: Algorithms, Data Structures, Embedded Systems, High-Performance Computer Architecture
- **Extracurriculars**: Association for Computing Machinery, HawkTrade, AWEA, Intramurals

Technical Skills

Languages: C++, C, Java, Python, Ruby, JavaScript, MATLAB, C#, LabVIEW
Frameworks & Libraries: React, Django, PyTorch, Node.js, Rails
Other: Git, Subversion, AWS, Terraform, Unity, Linux, Docker, CI/CD, Jira, Jama, TDD/BDD

Work Experience

Software Engineer II January 2026 – Present
Collins Aerospace *Cedar Rapids, IA*

- Developed embedded C/C++ software for cockpit communication and aircraft radio interfaces
- Authored software requirements and detailed designs to guide implementation and maintain traceability
- Developed and executed verification tests to validate functionality and requirements compliance

Software Engineer Intern May 2023 - December 2025
Collins Aerospace *Cedar Rapids, IA*

- Developed high-level embedded C++ software for cockpit user interfaces enabling pilot communication with ground stations
- Implemented backend C++ functionality to support communication system operations and interface behavior

Teaching Assistant August 2022 - May 2023
University of Iowa *Iowa City, IA*

- Served as a Teaching Assistant for the Computers in Engineering course with Dr. Gary Christensen and Dr. Yu-Jung Chu
- Served as a Teaching Assistant for the Introduction to Engineering Computing course with Dr. Justin Garvin and Dr. Mona Garvin
- Supported students through office hours, graded coursework, and taught advanced C++ in class and discussions

Projects

Streamline Bar App | *React, Django, Flutter, AWS, Scrum, Jira*

- Built a full-stack restaurant and bar management system with a React frontend, Django backend, and Flutter mobile app.
- Implemented mobile ordering, role-based user access, real-time order tracking, and analytics using AWS and RESTful APIs.

Summit Slope | *C#, Unity*

- Developed a medieval VR game for Meta Quest 2 where players run uphill, dodging obstacles and collecting power-ups
- Implemented stick-based movement controls, dynamic difficulty modes, and a real-time scoreboard to track fastest times and high scores

Leadership

Scrum Master & Project Engineering
Collins Aerospace, University of Iowa

- Led sprint planning and stand-ups, coordinated team priorities, and performed delegated PE duties including bid preparation and project planning